Metals & Non Metals

1) Physical prop.

Act. 1.

Fe, Cu, Mg & AL

→ Pure State → Shiny Surface

Act. 2.

→ Cut w. Knife
→ Hard.

Act. 3.

→ Fe, Pb & Cu

→ Beat → Thin Sheets
→ Prop. of Metals to draw into thin Sheets is called Malleability

Act 4

→ Fe, AL & Cu

→ Beat → Wires
→ Ductility

Ques - 5

→ Good Conductor of heat & electricity

Ionic Bond

→ Between Metal & Non Metal

Low Electrons — Gain Electrons

Metals → Outermost Shell mud contain 1, 2 or 3 electrons

When lose e. form cation

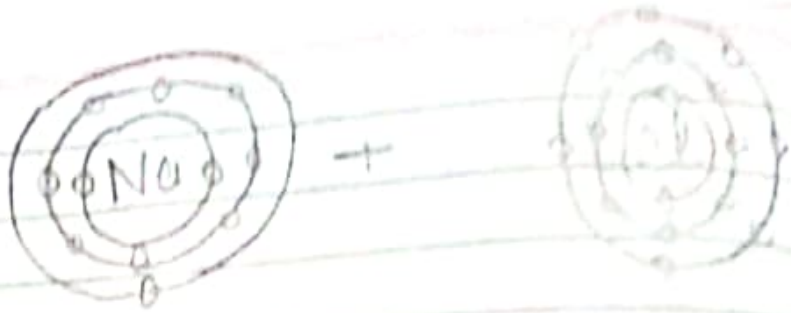
Non Metals → Gain e → Anion

* Between anion & Cation → Strong forces of attraction
  → Electric force of Attraction

eg. between Na^+ & Cl^-

• Formation OF Sodium Chloride

Na - 11	Cl - 17
2, 8, 1	2 8 8 7
↓	↓
$Na^+ \rightarrow e^-$	$Cl \pm \rightarrow e$
2, 8	2, 8, 8
Na^+	$:\ddot{\underset{\cdot\cdot}{Cl}}:$



→ NaCl.

• Bond formation in $MgCl_2$

$\underset{2,8,2}{Mg} \rightarrow \underset{2,8}{Mg^{+}} \rightarrow 2e^{-}$

$\underset{2,8,7}{Cl} \rightarrow + 2e \rightarrow 2Cl$

$= Mg^{2+} Cl_2$

• properties of ionic compounds.

→ Solid & crystalline (NaCl → cubic lattice)

→ M.P & B.P → High.

→ Soluble in H_2O

→ Do not conduct electricity in Solid State

→ Conduct in ~~Solid~~ Aq & Molten